Margaret Cancer Centre, University Health Network, Toronto, ON, Canada, [12]Department of Medical Biophysics, University of Toronto, Toronto, ON, Canada

Purpose/Objective(s): Magnetic resonance image guided radiotherapy (MRIgRT) of glioblastoma (GBM) is a promising treatment approach to monitor and adapt to dynamic tumor changes during radiation. These dynamics (tumor size, position, and morphology) during chemoradiation are inadequately understood, hindering the clinical implementation of MRIgRT. The purpose of this study was to quantify target changes during standard 6-week chemoradiation.

Materials/Methods: Sixty-one patients, treated with standard 60 Gy in 30 fractions with concurrent and adjuvant temozolomide, were prospectively imaged with a gadolinium-enhanced T1-weighted volumetric sequence (T1c) and volumetric T2/FLAIR at treatment planning (Fx 0), fraction 10 (Fx 10), fraction 20 (Fx 20), and one month post radiotherapy (P1M). All 61 patients completed imaging at Fx 0, Fx 10, and Fx 20, and 54 (89%) of these patients went on to complete imaging at P1M. Gross tumor volumes (GTV) and clinical target volumes (CTV; 1.5 cm expansion of the GTV respecting anatomical boundaries) were contoured at all timepoints. GTV and CTV dynamics were quantified by their absolute volume, volume relative to Fx 0 (V_{rel}), and the migration distance ($d_{migrate}$) defined as the maximum linear distance between a non-Fx 0 target and the respective Fx 0 target volume.

Results: The median GTV volume was 18.4 cm^3 (range: 1.1−110.5 cm^3), 14.7 cm^3 (0.9-115.1 cm^3), 13.7 cm^3 (0.6-174.2 cm^3), and 13.0 cm^3 (0.9-76.3 cm^3) at Fx 0, Fx 10, Fx 20, and P1M, respectively. The resulting median GTV V_{rel} was 0.88 (range: 0.35-1.40), 0.77 (0.09-2.13), and 0.71 (0.13-3.80) at Fx 10, Fx 20, and P1M, respectively. Based on these GTV contours, the corresponding median CTV volumes were 114.6 cm^3 (33.9−374.1 cm^3), 106.7 cm^3 (29.9-370.5 cm^3), 99.3 cm^3 (27.5-401.2 cm^3), and 101.7 cm^3 (30.3-376.0 cm^3) at Fx 0, Fx 10, Fx 20, and P1M, respectively, with ensuing median CTV V_{rel} at the non-Fx 0 timepoints of 0.96 (0.59-1.37), 0.88 (0.41-1.63), and 0.86 (0.39-2.38). The GTV $d_{migrate}$ was greater than 5 mm in 46%, 50%, and 54% of patients at Fx 10, Fx 20, and P1M, respectively. Similarly, $d_{migrate}$ of the CTV was greater than 5 mm in 54%, 54%, and 59% of patients. Extensive morphological changes were identified during chemoradiation even as target volumes decreased; among all patients, 32% and 38% had a decreasing GTV volume (i.e. $V_{rel} \leq 1$) with $d_{migrate} > 5$ mm at Fx 10 and Fx 20, respectively.

Conclusion: Highly variable and patient specific tumor changes during radiotherapy were observed supporting the need for routine MRI monitoring and highlights the potential need for adaptive treatment planning in this population.

Author Disclosure: **J. Stewart**: None. **A. Sahgal**: Research Grant; Elekta AB. Honoraria; Elekta AB, Varian. Consultant; Varian. Travel Expenses; Elekta AB, Varian. **Y. Lee**: None. **H. Soliman**: None. **C. Tseng**: None. **J. Detsky**: None. **Z.A. Husain**: None. **L. Ho**: None. **P. Maralani**: None. **G. Stanisz**: None. **J. Perry**: None. **J. Xie**: None. **M. Campbell**: None. **A. Lau**: None. **M.E. Ruschin**: Employee; Princess Margaret Cancer Centre. Patent/License Fees/Copyright; Gamma Knife ICON, Precision X-Ray SMART. **S.D. Myrehaug**: Research Grant; Novartis AG. Honoraria; Novartis AG, Ipsen.

3737

Withdrawn

3738

Low-Dose Whole Brain Radiation Therapy For Early Alzheimer's Dementia: Early Results From A Phase IIa Trial

C.L. Rogers,[1] S. Lageman,[2] T.J. Harris,[3] T. Thomas,[4] P. Boling,[2] J. Karis,[5] J. Shi,[5] and J. Fontanesi[6]; *[1]Barrow Neurological Institute, St. Joseph's Hospital and Medical Center, Phoenix, AZ, [2]Virginia Commonwealth University, Richmond, VA, [3]Virginia Commonwealth University, Richmond, VA, [4]Barrow Neurological Institute, Phoenix, AZ, [5]Barrow Neurological Institute, Phoenix, AZ, [6]William Beaumont Health Systems, Farmington Hills, MI*

Purpose/Objective(s): Report neurocognitive, imaging, and safety outcomes from a phase IIa trial of low-dose whole brain RT (LD-WBRT) for early Alzheimer's dementia (eAD) in 5 patients who completed all study assessments.

Materials/Methods: Based on improvements in amyloid burden and cognition in murine models and favorable outcomes with low-dose RT for systemic amyloidosis, we launched a human trial for eAD, defined by NINCDS-ADRDA criteria with confirmatory Florbetapir and FDG PET findings. The capacity to complete neurocognitive function (NCF), psychological function (PF), and quality of life (QOL) assessments was requisite. NCF was measured via a battery of validated assessments (WTAR, MMSE-2, HVLT-R, BVMT-R, WAIS-IV Digit Span and Coding, Trailmaking Parts A+B, COWAT, Semantic Fluency, Grooved Pegboard, BNT and JLO), PF via PHQ-9 and GAD-7, and QOL per QOL-AD and QUALID. Participant and informant ratings of cognition consisted of ECOG questionnaires. Participants underwent screening physical examination, screening as well as pre-treatment NCF, PF and QOL assessments, and screening Florbetapir and FDG PET imaging. Each received LD-WBRT (2Gyx5). Post-treatment evaluations included 6wk, 3mo, 9mo, and 12mo physical exams, NCF, PF, and QOL assessments (with BNT and JLO only at baseline and 12mo), 6mo FDG and Florbetapir PET, and Data Safety and Monitoring Committee (DSMC) toxicity evaluation.

Results: 5 patients were treated with LD-WBRT and completed all study assessments. Mean age was 73.2 (range 69-77) yrs, 3 female and 2 male. Comparing pre-treatment and 12mo MMSE-2 T-scores, 3 improved, 1 remained stable, and 1 declined. Skills expected to wane in early AD (naming, learning and memory) remained broadly stable at 1 year in 4 of 5, with trends for improved verbal learning in 3. The participant who declined in these measures and on Grooved Pegboard and Semantic Fluency, met the lowest-end inclusion criteria and was the eldest. Processing speed (Grooved Pegboard, WAIS-IV Coding, Trailmaking A), attention (WAIS-IV Digit Span), visuospatial skills (JLO, BVMT-R Copy), executive functions (Trailmaking A+B, lexical fluency), mood (PHQ-9, GAD-7) and QOL (QOL-AD, QUALID, ECOG) assessments remained stable, with exception of some declines in semantic fluency suggestive of temporal lobe dysfunction. However, there were improvements in several domains at 9 to 12mo. PET imaging remained stable with possible modest improvements in hippocampal amyloid and parietotemporal FDG. Results were independently reviewed by the Data Safety monitoring committee, and no safety issues encountered except transient epilation.

Conclusion: 4 of the initial 5 patients with eAD treated with LD-WBRT experienced overall stability in NCF, PF, and QOL, with trends for improved verbal learning, and stability to possible improvement in PET imaging. The trial will continue to 15 patients at 2Gy x 5, then 15 at 2Gy x 10. Longer follow up is now planned to evaluate whether improvements in several domains at 9 to 12 months are confirmed.

Author Disclosure: **C. Rogers**: Research Grant; NCI - ActNOW. **S. Lageman**: None. **T.J. Harris**: None. **T. Thomas**: None. **P. Boling**: None. **J. Karis**: None. **J. Shi**: None. **J. Fontanesi**: Independent Contractor; William Beaumont Hospital.

3739

Late Midterm Outcomes of Uveal Melanoma Treated with Intraoperative Ultrasound Guided I^{125} Brachytherapy Using Custom Built Eye Plaques

A.J. Wong,[1] B.S. Teh,[2] B.T. Nguyen,[3] R. Pino,[2] M.E. Bretana,[4] E. Bernicker,[2] P. Chevez-Barrios,[2] E.B. Butler,[2] and A. Schefler[2,4]; *[1]The University of Texas Medical Branch, Galveston, TX, [2]Houston Methodist Hospital, Houston, TX, [3]Arizona College of Osteopathic Medicine at Midwestern University, Glendale, AZ, [4]Retina Consultants of Houston, Houston, TX*

Purpose/Objective(s): Uveal melanoma is the most common histology of primary intraocular tumors. Plaque brachytherapy is the most common treatment modality in the United States. We aim to demonstrate that image guided brachytherapy with intraoperative ultrasound at time of plaque placement is associated with increased local control.